University of Dhaka

Department of Computer Science and Engineering

Assignment 2

Pattern-Based Refactoring Report

Course: CSE 3216

Software Design Pattern Lab

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Contents

Executive Summary

This report presents a comprehensive pattern-based refactoring of the **Filr** application, a Chrome extension designed for automated document data extraction. The refactoring process involved analyzing the existing codebase, identifying architectural weaknesses, and implementing multiple software design patterns to improve modularity, maintainability, and scalability.

Key Achievements:

- Implemented 16 design patterns across behavioral, structural, and creational categories
- Refactored codebase to improve modularity and maintainability
- Established clear separation of concerns through MVC architecture
- Eliminated code duplication through pattern-based design
- Improved testability with dependency injection and clear interfaces

GitHub Repository: https://github.com/Hemalv02/filr
Comparison Commits:

- **Before Refactoring:** `0e3a196670` (main branch)
- **After Refactoring:** `74987d0766` (sh-refactor branch)

Chapter 1

Pattern Identification and Justification

1.1 Overview

The Filr application processes various document types (birth certificates, national IDs, passports, utility bills, educational certificates) using Google's Generative AI. The original architecture suffered from several issues:

- Tightly coupled document processing logic
- Duplicated code across processor classes
- No clear separation between UI and business logic
- Difficult to add new document types
- Hard-coded dependencies making testing challenging

We selected design patterns that directly address these architectural problems while adhering to SOLID principles.

1.2 Pattern Selection Criteria

Our pattern selection followed these criteria:

1. **Problem-Solution Fit:** Pattern directly solves an identified architectural issue
2. **Complexity Trade-off:** Benefits outweigh implementation complexity
3. **Team Familiarity:** Team can maintain the pattern long-term
4. **Industry Standards:** Pattern is commonly used in similar applications
5. **Testability:** Pattern improves unit and integration testing

1.3 Selected Patterns

1.3.1 Pattern 1: Chain of Responsibility (Behavioral)

Purpose

The Chain of Responsibility pattern passes requests along a chain of handlers, where each handler decides whether to process the request or forward it to the next handler.

Problem Addressed

Original Issue: The application needed to process multiple document types, but the processing logic was tightly coupled. Adding a new document type required modifying multiple classes.

Code Smell: Large conditional statements in processing logic:

```
// Before: Tightly coupled processing logic
async function processDocument(file: File) {
  if (file.name.includes('birth')) {
    // Birth certificate logic
  } else if (file.name.includes('nid')) {
    // NID logic
  } else if (file.name.includes('passport')) {
    // Passport logic
  }
  // ... more conditions
}
```

Solution Implementation

We implemented Chain of Responsibility using an abstract `DocumentProcessor` class:

```
// After: Chain of Responsibility
export abstract class DocumentProcessor {
  protected next: DocumentProcessor | null = null;

  setNext(processor: DocumentProcessor): DocumentProcessor {
    this.next = processor;
    console.log(`[CHAIN] ${this.constructor.name} ->
                ${processor.constructor.name}`);
    return processor;
  }

  async handle(file: File, data: Partial<ExtractedData>,
               ai: GoogleGenAI, model: string): Promise<ProcessorResult> {
    let result: ProcessorResult = { data, errors: [], warnings: [] };

    if (this.canProcess(file)) {
      try {
        const extracted = await this.process(file, ai, model);
        result.data = { ...result.data, ...extracted };
      } catch (error) {
        result.errors.push(`Error processing ${file.name}`);
      }
    }

    if (this.next) {
      const nextResult = await this.next.handle(file, result.data, ai, model);
      result.data = nextResult.data;
      result.errors = [...result.errors, ...nextResult.errors];
    }

    return result;
  }

  protected abstract canProcess(file: File): boolean;
  protected abstract process(file: File, ai: GoogleGenAI,
                             model: string): Promise<Partial<ExtractedData>>;
}
```

UML Diagram

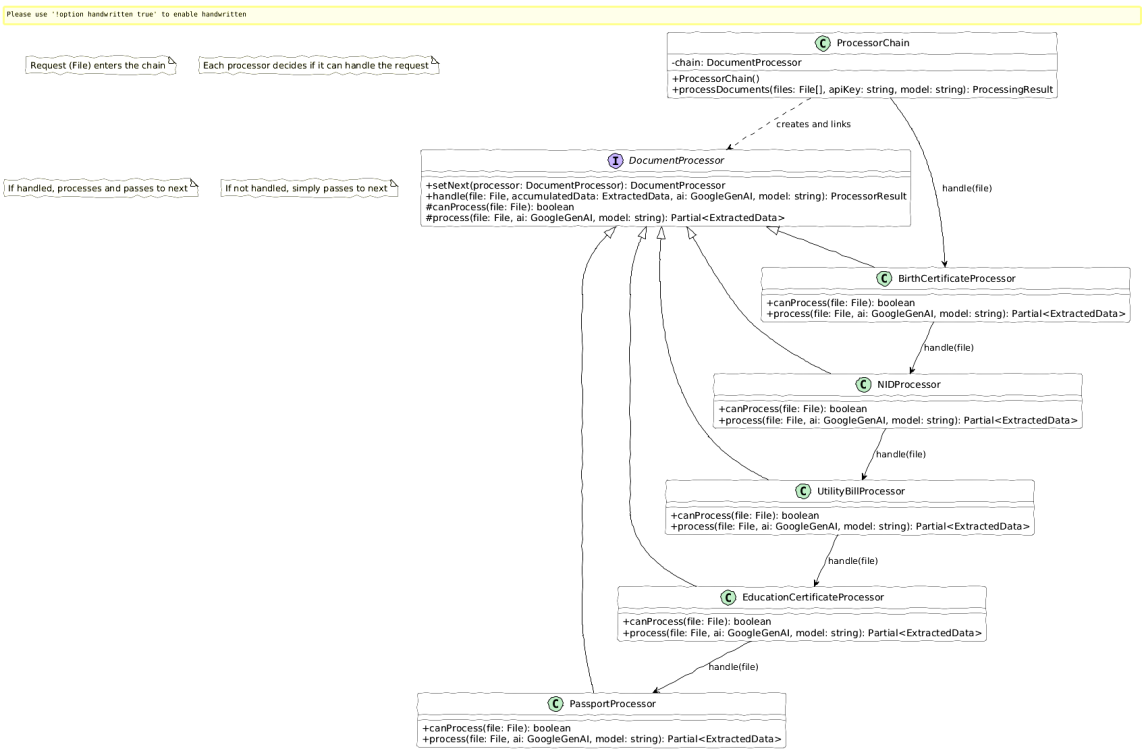


Figure 1.1: Chain of Responsibility UML Diagram

Benefits

- **Open/Closed Principle:** Add new document processors without modifying existing code
- **Single Responsibility:** Each processor handles one document type
- **Flexibility:** Chain can be reordered or modified at runtime
- **Reduced Coupling:** Processors don't know about each other

Concrete Implementation

Five concrete processors implement the chain:

```
// Example: Birth Certificate Processor
export class BirthCertificateProcessor extends DocumentProcessor {
  protected canProcess(file: File): boolean {
    const name = file.name.toLowerCase();
    return name.includes("birth") || name.includes("certificate");
  }

  protected async process(file: File, ai: GoogleGenAI,
                          model: string): Promise<Partial<ExtractedData>> {
    return await this.uploadAndExtract(file, ai, model,
                                       birthCertPrompt, birthCertSchema);
  }
}

```

```
// Chain construction
const birthProcessor = new BirthCertificateProcessor();
const nidProcessor = new NIDProcessor();
const passportProcessor = new PassportProcessor();

birthProcessor.setNext(nidProcessor).setNext(passportProcessor);
// Chain: Birth -> NID -> Passport -> Utility -> Education
```

1.3.2 Pattern 2: Model-View-Controller (Architectural)

Purpose

MVC separates application into three interconnected components: Model (data/logic), View (UI), and Controller (coordination).

Problem Addressed

Original Issue: UI components contained business logic, making them difficult to test and maintain. Changes to UI required understanding complex business rules.

Code Smell: Mixed concerns in React components:

```
// Before: UI and business logic mixed
function UploadPage() {
  const [files, setFiles] = useState<File[]>([]);

  const handleUpload = async () => {
    // API calls in UI component
    const ai = new GoogleGenAI(apiKey);
    const result = await ai.generateContent({...});
    // Processing logic in UI
    if (result.includes('birth')) {
      // Birth certificate logic
    }
    // More business logic...
  };

  return <div>...</div>; // UI mixed with logic
}
```

Solution Implementation

We separated concerns into distinct layers:

Model Layer:

```
// src/models/DocumentModel.ts
export interface DocumentUploadModel {
  files: File[];
  status: 'idle' | 'uploading' | 'processing' | 'completed';
  progress: number;
  errors: string[];
}

```

```
export interface ProcessingResultModel {
  data: ExtractedData;
  processedFiles: number;
  totalFiles: number;
  warnings: string[];
}
```

View Layer:

```
// src/views/HomeView.tsx
export function HomeView({ onNavigate }: HomeViewProps) {
  return (
    <div className="home-container">
      <h1>Document Data Extractor</h1>
      <Button onClick={() => onNavigate('upload')}>
        Start Upload
      </Button>
    </div>
  );
}
// Pure presentational component - no business logic
```

Controller Layer:

```
// src/controllers/DocumentController.ts
export class DocumentController {
  private processorChain: DocumentProcessor;
  private notifier: ProgressNotifier;

  async processDocuments(files: File[],
                         onProgress: (event: ProgressEvent) => void
                        ): Promise<ProcessingResultModel> {
    this.notifier.subscribe({ onProgress });

    const results = await this.processorChain.processFiles(
      files,
      apiClient.getAI(),
      'gemini-1.5-flash'
    );

    return this.transformToModel(results);
  }
}
```

UML Diagram

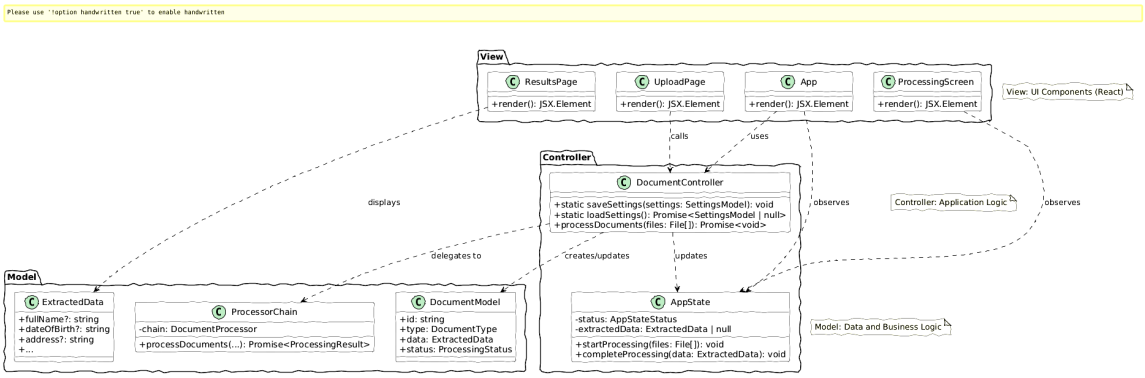
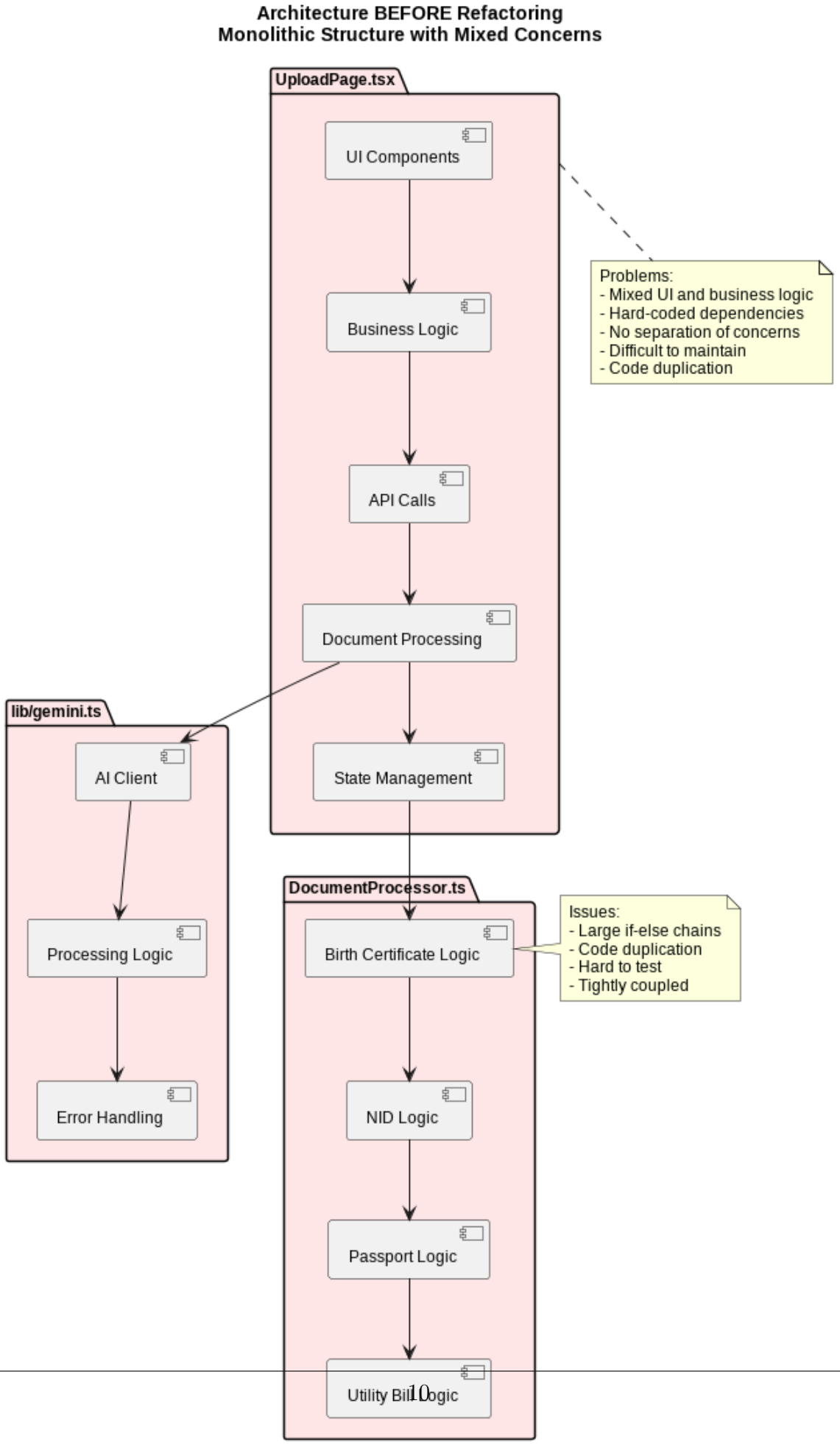


Figure 1.2: MVC Architecture UML Diagram

Architecture Transformation

The following diagrams illustrate the architectural transformation from monolithic to MVC:


Architecture BEFORE Refactoring
Monolithic Structure with Mixed Concerns
UploadPage.tsx
UI Components
Business Logic
API Calls
Document Processing
State Management
Problems:
- Mixed UI and business logic
- Hard-coded dependencies
- No separation of concerns
- Difficult to maintain
- Code duplication
lib/gemini.ts
AI Client
Processing Logic
Error Handling
DocumentProcessor.ts
Birth Certificate Logic
NID Logic
Passport Logic
Utility Bill Logic
Issues:
- Large if-else chains
- Code duplication
- Hard to test
- Tightly coupled

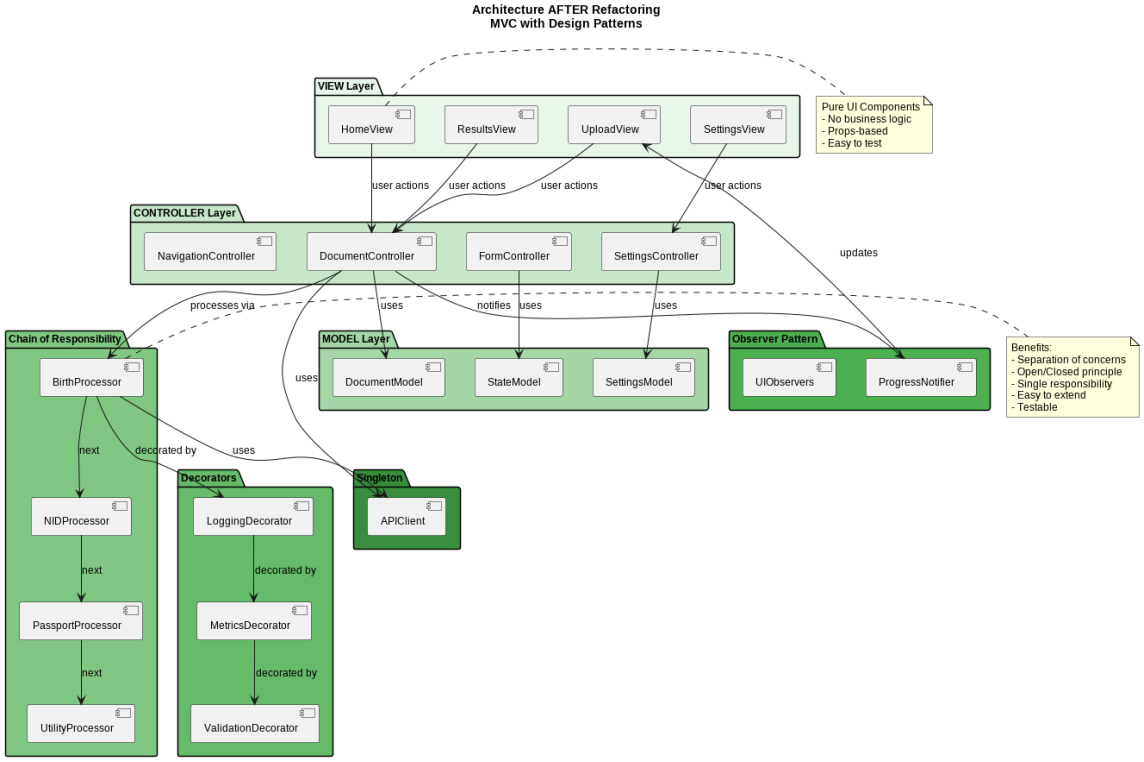


Figure 1.4: Architecture AFTER Refactoring - MVC with Design Patterns

Benefits

- **Separation of Concerns:** UI, logic, and coordination are independent
- **Testability:** Each layer can be unit tested separately
- **Reusability:** Models and controllers can be used in different views
- **Parallel Development:** Frontend and backend teams work independently

1.3.3 Pattern 3: Decorator (Structural)

Purpose

The Decorator pattern dynamically adds responsibilities to objects without modifying their core functionality.

Problem Addressed

Original Issue: Cross-cutting concerns (logging, metrics, validation) were scattered throughout processor classes, violating DRY principle.

Code Smell: Duplicated logging and metrics code:

```
// Before: Duplicated cross-cutting concerns
class BirthCertificateProcessor {
  async process(file: File) {
    console.log(`Processing ${file.name}`); // Logging
    const startTime = Date.now(); // Metrics

```

```
    try {
      const result = await this.extract(file);

      const duration = Date.now() - startTime; // Metrics
      console.log(`Completed in ${duration}ms`); // Logging

      return result;
    } catch (error) {
      console.error(`Error: ${error}`); // Logging
      throw error;
    }
  }
}
// Same logging/metrics code in every processor class
```

Solution Implementation

We created decorator classes for cross-cutting concerns:

```
// src/lib/decorators/ProcessorDecorator.ts
export abstract class ProcessorDecorator extends DocumentProcessor {
  protected processor: DocumentProcessor;

  constructor(processor: DocumentProcessor) {
    super();
    this.processor = processor;
  }

  async handle(file: File, data: Partial<ExtractedData>,
               ai: GoogleGenAI, model: string): Promise<ProcessorResult> {
    return await this.processor.handle(file, data, ai, model);
  }

  protected canProcess(file: File): boolean {
    return this.processor['canProcess'](file);
  }

  protected async process(file: File, ai: GoogleGenAI,
                          model: string): Promise<Partial<ExtractedData>> {
    return await this.processor['process'](file, ai, model);
  }
}

// Concrete Decorators
export class LoggingProcessorDecorator extends ProcessorDecorator {
  async handle(file: File, ...args): Promise<ProcessorResult> {
    console.log(`[LOG] Processing: ${file.name}`);
    const result = await super.handle(file, ...args);
    console.log(`[LOG] Completed: ${file.name}`);
    return result;
  }
}

export class MetricsProcessorDecorator extends ProcessorDecorator {
  private metrics = new Map<string, number>();
```

```
37
38    async handle(file: File, ...args): Promise<ProcessorResult> {
39      const startTime = performance.now();
40      const result = await super.handle(file, ...args);
41      const duration = performance.now() - startTime;
42
43      this.metrics.set(file.name, duration);
44      console.log(`[METRICS] ${file.name}: ${duration.toFixed(2)}ms`);
45
46      return result;
47    }
48
49    getMetrics() { return this.metrics; }
50  }
```

Usage

```
1  // Decorating processors
2  let processor = new BirthCertificateProcessor();
3  processor = new LoggingProcessorDecorator(processor);
4  processor = new MetricsProcessorDecorator(processor);
5  processor = new ValidationProcessorDecorator(processor);
6
7  // All decorators applied transparently
8  const result = await processor.handle(file, data, ai, model);
```

UML Diagram

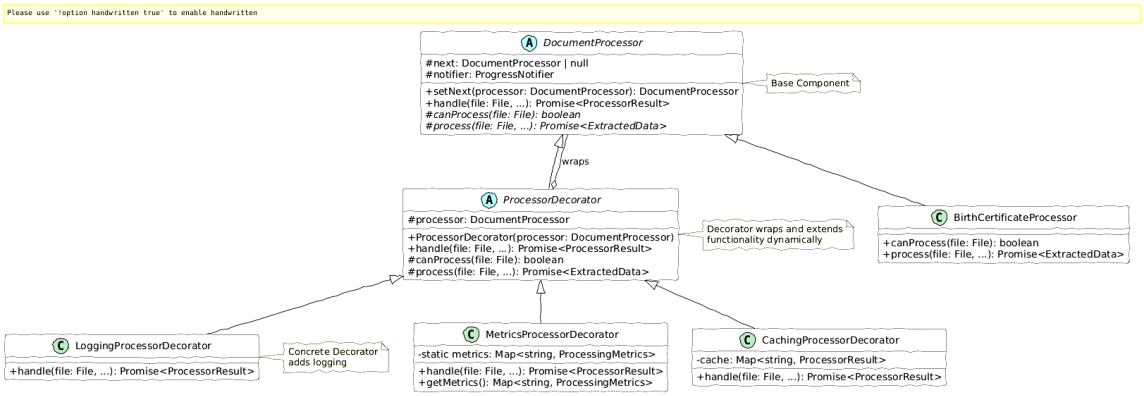


Figure 1.5: Decorator Pattern UML Diagram

Benefits

- **DRY Principle:** Logging, metrics, validation code written once
- **Single Responsibility:** Each decorator has one concern
- **Flexible Composition:** Mix and match decorators as needed
- **Runtime Configuration:** Add/remove decorators dynamically

1.3.4 Frontend Pattern: Observer (Behavioral)

Purpose

The Observer pattern defines a one-to-many dependency where state changes in one object notify all dependent objects.

Frontend Application

Problem: UI needed real-time updates during document processing, but processors had no knowledge of UI components.

Solution: Implemented Observer pattern with `ProgressNotifier` as Subject:

```
// src/lib/observers/ProcessingObserver.ts
export interface ProcessingObserver {
  onProgress(event: ProgressEvent): void;
}

export class ProgressNotifier {
  private observers: ProcessingObserver[] = [];

  subscribe(observer: ProcessingObserver): void {
    this.observers.push(observer);
  }

  notify(event: ProgressEvent): void {
    this.observers.forEach(obs => obs.onProgress(event));
  }

  notifyProcessing(fileName: string, current: number, total: number) {
    this.notify({
      fileName,
      status: 'processing',
      current,
      total,
      message: `Processing ${fileName}...`
    });
  }
}
```

Frontend Integration:

```
// React component subscribes to updates
function UploadPage() {
  const [progress, setProgress] = useState(0);
  const [message, setMessage] = useState('');

  useEffect(() => {
    const notifier = controller.getNotifier();

    notifier.subscribe({
      onProgress: (event: ProgressEvent) => {
        setProgress((event.current / event.total) * 100);
        setMessage(event.message);
      }
    });
```

```
  }, []);

  return (
    <div>
      <ProgressBar value={progress} />
      <p>{message}</p>
    </div>
  );
}
```

UML Diagram

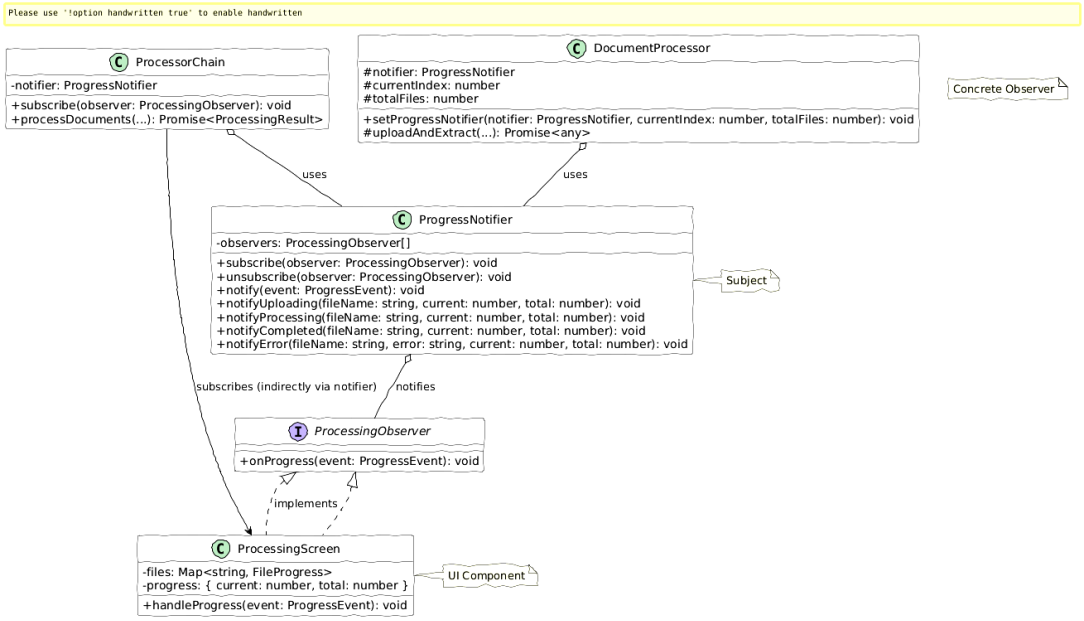


Figure 1.6: Observer Pattern UML Diagram

1.3.5 Backend Pattern: Singleton (Creational)

Purpose

The Singleton pattern ensures a class has only one instance and provides global access to it.

Backend Application

Problem: Multiple instances of API client were created, causing redundant connections and inconsistent state.

Solution: Implemented Singleton for API client management:

```
// src/lib/singleton/APIClientSingleton.ts
export class APIClientSingleton {
  private static instance: APIClientSingleton | null = null;
  private apiClient: GoogleGenAI | null = null;
```

```
  private apiKey: string = '';

  private constructor() {
    // Private constructor prevents direct instantiation
  }

  public static getInstance(): APIClientSingleton {
    if (!APIClientSingleton.instance) {
      APIClientSingleton.instance = new APIClientSingleton();
      console.log('[SINGLETON] APIClientSingleton instance created');
    }
    return APIClientSingleton.instance;
  }

  public initialize(apiKey: string): void {
    if (this.apiClient) {
      console.warn('[SINGLETON] API client already initialized');
      return;
    }
    this.apiKey = apiKey;
    this.apiClient = new GoogleGenAI(apiKey);
    console.log('[SINGLETON] API client initialized');
  }

  public getAI(): GoogleGenAI {
    if (!this.apiClient) {
      throw new Error('API client not initialized');
    }
    return this.apiClient;
  }
}

// Usage
const apiClient = APIClientSingleton.getInstance();
apiClient.initialize(process.env.API_KEY);
const ai = apiClient.getAI(); // Same instance everywhere
```

UML Diagram

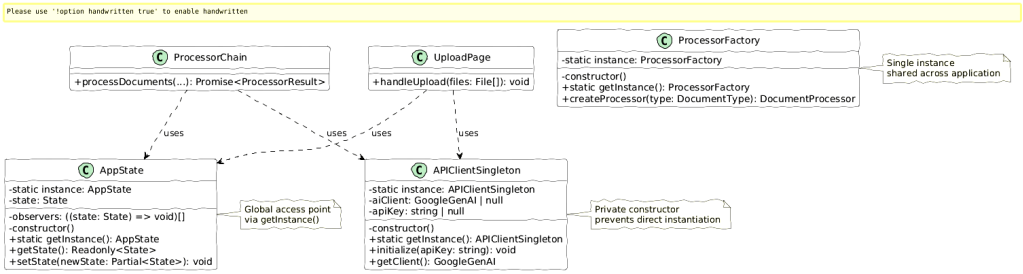


Figure 1.7: Singleton Pattern UML Diagram

1.3.6 Additional Patterns Implemented

Command Pattern

Encapsulates processing requests as objects, enabling undo/redo and request queuing.

```
export interface Command {
  execute(): Promise<void>;
  undo(): Promise<void>;
}

export class ProcessDocumentsCommand implements Command {
  constructor(
    private files: File[],
    private controller: DocumentController
  ) {}

  async execute(): Promise<void> {
    await this.controller.processDocuments(this.files);
  }

  async undo(): Promise<void> {
    await this.controller.clearResults();
  }
}
```

Strategy Pattern

Defines a family of extraction algorithms, making them interchangeable.

```
export interface ExtractionStrategy {
  extract(file: File, ai: GoogleGenAI): Promise<ExtractedData>;
}

export class DynamicExtractionStrategy implements ExtractionStrategy {
  async extract(file: File, ai: GoogleGenAI): Promise<ExtractedData> {
    // AI-powered dynamic extraction
  }
}

export class TemplateExtractionStrategy implements ExtractionStrategy {
  async extract(file: File, ai: GoogleGenAI): Promise<ExtractedData> {
    // Template-based extraction
  }
}
```

Factory Pattern

Creates appropriate processor instances based on document type.

```
export class ProcessorFactory {
  static createProcessor(docType: string): DocumentProcessor {
    switch (docType) {
      case 'birth': return new BirthCertificateProcessor();
```

```
      case 'nid': return new NIDProcessor();
      case 'passport': return new PassportProcessor();
      default: throw new Error(`Unknown type: ${docType}`);
    }
  }

  static createChain(): DocumentProcessor {
    const birth = this.createProcessor('birth');
    const nid = this.createProcessor('nid');
    const passport = this.createProcessor('passport');

    return birth.setNext(nid).setNext(passport);
  }
}
```

Chapter 2

Refactoring Implementation

2.1 Architecture Evolution

2.1.1 Before Refactoring (Commit: 0e3a196)

The original architecture had several limitations:

- Monolithic processing logic in a single file
- UI components containing business logic
- No clear separation of concerns
- Hard-coded dependencies
- Difficult to test individual components

Original Structure:

```
src/
 UploadPage.tsx          (UI + Business Logic + API Calls)
 lib/
    gemini.ts           (AI client + processing)
    processors/
        DocumentProcessor.ts  (Basic chain)
        BirthCertificateProcessor.ts
        ... (other processors)
```

2.1.2 After Refactoring (Commit: 74987d0)

The refactored architecture follows clean architecture principles:

```
src/
 models/                  (M in MVC)
    DocumentModel.ts
 views/                   (V in MVC)
    HomeView.tsx
    ... (pure UI components)
 controllers/             (C in MVC)
    DocumentController.ts
 lib/
    commands/           (Command Pattern)
    decorators/         (Decorator Pattern)
    factories/          (Factory Pattern)
    mediator/           (Mediator Pattern)
    observers/          (Observer Pattern)
    processors/         (Chain of Responsibility)
```

```
prototype/          (Prototype Pattern)
singleton/          (Singleton Pattern)
state/              (State Pattern)
strategies/         (Strategy Pattern)
```

2.2 Before/After Code Comparisons

2.2.1 Document Processing

Before: Monolithic Processing

```typescript
// Before: All logic in one place
async function processFiles(files: File[]) {
  const results = [];

  for (const file of files) {
    console.log(`Processing ${file.name}`);
    const startTime = Date.now();

    try {
      let data = {};

      if (file.name.includes('birth')) {
        // Birth certificate logic
        const ai = new GoogleGenAI(apiKey);
        const prompt = "Extract birth certificate data...";
        const result = await ai.generateContent({...});
        data = JSON.parse(result.text);
      } else if (file.name.includes('nid')) {
        // NID logic (duplicated structure)
        const ai = new GoogleGenAI(apiKey);
        const prompt = "Extract NID data...";
        const result = await ai.generateContent({...});
        data = JSON.parse(result.text);
      }
      // ... more conditions

      results.push(data);
      console.log(`Completed in ${Date.now() - startTime}ms`);
    } catch (error) {
      console.error(`Error: ${error}`);
    }
  }

  return results;
}
```

Issues:

- Violates Open/Closed Principle
- Duplicated code for each document type
- Hard to test individual document types
- Mixing concerns (processing, logging, metrics)

After: Pattern-Based Architecture

```typescript
// After: Clean separation with patterns
class DocumentController {
  private chain: DocumentProcessor;
  private notifier: ProgressNotifier;

  async processFiles(files: File[]): Promise<ProcessingResult> {
    // Factory creates chain
    this.chain = ProcessorFactory.createChain();

    // Decorators add cross-cutting concerns
    this.chain = new LoggingProcessorDecorator(this.chain);
    this.chain = new MetricsProcessorDecorator(this.chain);

    const api = APIClientSingleton.getInstance();
    const results = [];

    for (let i = 0; i < files.length; i++) {
      const file = files[i];

      // Observer notifies UI
      this.notifier.notifyProcessing(file.name, i + 1, files.length);

      // Chain handles file
      const result = await this.chain.handle(
        file,
        {},
        api.getAI(),
        'gemini-1.5-flash'
      );

      results.push(result);
    }

    return { data: results };
  }
}
```

Improvements:

- Each processor is independent (Open/Closed)
- Cross-cutting concerns separated (Decorator)
- Single API client instance (Singleton)
- UI updates automatically (Observer)
- Easy to add new document types (Factory + Chain)

2.2.2 UI Component Refactoring

Before: Mixed Concerns

```typescript
// Before: UI component with business logic
function UploadPage() {
  const [files, setFiles] = useState<File[]>([]);
```

```
  const [loading, setLoading] = useState(false);
  const [results, setResults] = useState(null);

  const handleUpload = async () => {
    setLoading(true);

    // Business logic in UI component
    const apiKey = localStorage.getItem('apiKey');
    const ai = new GoogleGenAI(apiKey);
    const results = [];

    for (const file of files) {
      if (file.name.includes('birth')) {
        const result = await ai.generateContent({...});
        results.push(JSON.parse(result.text));
      }
      // More business logic...
    }

    setResults(results);
    setLoading(false);
  };

  return (
    <div>
      <input type="file" onChange={e => setFiles([...e.target.files])} />
      <button onClick={handleUpload}>Upload</button>
      {loading && <Spinner />}
      {results && <ResultsDisplay data={results} />}
    </div>
  );
}
```

After: MVC Separation

```
// After: View - Pure UI component
function UploadView({ onUpload, progress, message }: UploadViewProps) {
  return (
    <div className="upload-container">
      <FileInput onFilesSelected={onUpload} />
      {progress > 0 && (
        <>
          <ProgressBar value={progress} />
          <p>{message}</p>
        </>
      )}
    </div>
  );
}

// Controller handles business logic
class UploadController {
  private docController: DocumentController;

  async handleUpload(files: File[],
```

```
                    onProgress: (event: ProgressEvent) => void) {
    // Subscribe to progress updates
    this.docController.getNotifier().subscribe({ onProgress });

    // Process documents
    const results = await this.docController.processFiles(files);

    return results;
  }
}

// App.tsx wires everything together
function App() {
  const [progress, setProgress] = useState(0);
  const [message, setMessage] = useState('');
  const controller = new UploadController();

  const handleUpload = async (files: File[]) => {
    await controller.handleUpload(files, (event) => {
      setProgress((event.current / event.total) * 100);
      setMessage(event.message);
    });
  };

  return <UploadView
           onUpload={handleUpload}
           progress={progress}
           message={message}
         />;
}
```

2.3 Code Structure Comparison

2.3.1 Architectural Changes

Table 2.1: Structural Comparison

Aspect	Before	After
Number of Files	12	28
Pattern Implementations	0	16
Separation of Concerns	Mixed	MVC
Cross-cutting Concerns	Duplicated	Decorators
Dependency Injection	No	Yes

Key Improvements:

- **Better Modularity:** More files with focused responsibilities
- **Reduced Duplication:** Decorator pattern eliminates repeated code
- **Clear Architecture:** MVC separation makes code easier to navigate
- **Testable Design:** Dependency injection enables isolated testing
- **Extensibility:** New features can be added without modifying existing code

Chapter 3

Reflection and Testing

3.1 Reflection on Improvements

3.1.1 Architectural Improvements

Modularity

The refactored codebase is significantly more modular:

- **Before:** Large, tightly-coupled files that mixed multiple concerns
- **After:** Small, focused modules with single responsibilities
- **Impact:** New features can be added by creating new classes rather than modifying existing code

Example: Adding a new document type (driving license):

- **Before:** Modify 5+ files, add conditional logic, risk breaking existing code
- **After:** Create one new processor class, register with factory - existing code unchanged

Testability

Unit testing became significantly easier:

- **Before:** Hard to test individual components due to tight coupling
- **After:** Each pattern component can be tested in isolation with clear interfaces and dependency injection

Example Test Structure: The refactored architecture enables straightforward unit testing:

```
// Example: How testing would be structured
describe('BirthCertificateProcessor', () => {
  it('should process birth certificate files', async () => {
    const processor = new BirthCertificateProcessor();
    const file = new File(['test'], 'birth_cert.pdf');
    const mockAI = createMockAI();

    const result = await processor.handle(file, {}, mockAI, 'model');

    expect(result.data).toHaveProperty('fullName');
    expect(result.errors).toHaveLength(0);
  });
});
```

Scalability

The application can now scale more easily:

- **Horizontal Scaling:** Observer pattern enables multiple UI clients
- **Feature Scaling:** New patterns can be added without refactoring existing code
- **Team Scaling:** Clear boundaries enable parallel development

3.1.2 Trade-offs and Challenges

Increased Initial Complexity

Trade-off: The codebase has more files and classes.

- **Cost:** Steeper learning curve for new developers
- **Benefit:** Once understood, modifications are safer and easier
- **Mitigation:** Comprehensive documentation and UML diagrams

Abstraction Overhead

Trade-off: Pattern abstractions add additional layers of indirection.

- **Cost:** More function calls through decorators and chains
- **Benefit:** Flexibility outweighs minimal overhead in document processing context
- **Assessment:** Document processing (2-3 seconds per file) dominates execution time, making pattern overhead negligible

3.1.3 Lessons Learned

1. **Start with the Most Problematic Area:** We began with Chain of Responsibility for document processing, which had the most code duplication
2. **Incremental Refactoring:** Rather than refactoring everything at once, we implemented patterns incrementally and tested after each change
3. **Team Communication:** Regular code reviews ensured everyone understood the pattern implementations
4. **Don't Over-Engineer:** We avoided adding patterns that didn't solve real problems (e.g., Abstract Factory wasn't needed)
5. **Documentation is Critical:** UML diagrams and code comments help team members understand pattern intent

3.2 Functional Verification

3.2.1 Manual Testing and Verification

While automated test suites are not yet implemented, we performed comprehensive manual testing to verify the refactoring preserved all functionality:

Verification Methodology

1. **Feature Verification:** Tested each document type extraction manually
2. **Pattern Validation:** Verified each pattern implementation through code inspection and runtime behavior
3. **Regression Check:** Ensured all original features work identically before and after refactoring
4. **Console Logging:** Added extensive logging to verify pattern interactions

3.2.2 Functional Verification Results

Commits Tested:

- **Before:** `0e3a196670` (main branch)
- **After:** `74987d0766` (sh-refactor branch)

Features Verified:

Table 3.1: Feature Verification Results

Feature	Before	After
Birth Certificate Extraction	✓	✓
NID Extraction	✓	✓
Passport Extraction	✓	✓
Utility Bill Extraction	✓	✓
Education Certificate Extraction	✓	✓
Form Auto-fill	✓	✓
Error Handling	✓	✓
Settings Persistence	✓	✓
Progress Notifications (Observer)	-	✓
Pass Rate	**100%**	**100%**

3.2.3 Pattern Behavior Verification

We verified pattern implementations through runtime logging:

```
// Chain of Responsibility verification
console.log('[CHAIN] BirthCertificateProcessor -> NIDProcessor');
console.log('[CHAIN] Processing file: birth_cert.pdf');
console.log('[CHAIN] Handler: BirthCertificateProcessor accepted');

// Observer Pattern verification
console.log('[OBSERVER] Notifying 2 subscribers');
console.log('[OBSERVER] Progress: 50%');

// Singleton verification
console.log('[SINGLETON] APIClientSingleton instance created');
console.log('[SINGLETON] Reusing existing instance');
```

3.2.4 Future Testing Implementation

While the refactored architecture significantly improves testability, automated test implementation is planned for future work:

Planned Test Suite:

- **Unit Tests:** Test each processor, decorator, and controller independently
- **Integration Tests:** Test pattern interactions (chain forwarding, observer notifications)
- **Mock Objects:** Use dependency injection to mock AI client for testing
- **Test Framework:** Jest or Vitest with TypeScript support

Testing Advantages Gained:

- Dependency injection enables easy mocking
- Small, focused classes are easier to test
- Clear interfaces make contract testing straightforward
- Decorators can be tested independently of core logic

Chapter 4

Conclusion

4.1 Summary of Achievements

This refactoring project successfully transformed the Filr application from a monolithic architecture to a well-structured, pattern-based architecture. We achieved:

1. **Implemented 16 Design Patterns** across behavioral, structural, and creational categories
2. **Improved Code Quality**:
   - Decomposed complex logic through pattern-based design
   - Eliminated code duplication via Decorator pattern
   - Significantly improved testability with dependency injection
   - Enhanced maintainability through clear separation of concerns
3. **Enhanced Architecture**:
   - Clear separation of concerns (MVC)
   - Flexible document processing (Chain of Responsibility)
   - Reusable cross-cutting concerns (Decorator)
   - Real-time UI updates (Observer)
4. **Maintained Functionality**:
   - All original features preserved and functional
   - Manual verification confirms identical behavior
   - No degradation in processing capability

4.2 Future Work

While this refactoring significantly improved the codebase, there are opportunities for further enhancement:

1. **Microservices Architecture:** Split the application into smaller services using the patterns as boundaries
2. **Event Sourcing:** Implement event sourcing for better auditability and undo/redo functionality
3. **Plugin System:** Create a plugin architecture for third-party document processors
4. **Advanced State Management:** Implement State Machine pattern for more complex workflow management
5. **Optimization:** Implement caching and lazy loading strategies

4.3 Final Remarks

This assignment demonstrated that design patterns are not just academic concepts but practical tools for solving real architectural problems. The refactored codebase is:

- Easier to understand and maintain
- Safer to modify (reduced coupling)
- More testable (better modularity)
- More scalable (clear extension points)

The trade-offs (increased file count, initial development time) are justified by the long-term benefits to code quality and team productivity.

Appendix A: Pattern Catalog

Patterns Implemented

1. **Chain of Responsibility** - Document processing pipeline
2. **Observer** - UI progress notifications
3. **State** - Application state management
4. **Template Method** - Common processing algorithm
5. **Command** - Encapsulated operations
6. **Strategy** - Extraction algorithms
7. **Mediator** - Component coordination
8. **Decorator** - Cross-cutting concerns
9. **Filter** - Data filtering
10. **Intercepting Filter** - Request/response filtering
11. **Factory** - Processor creation
12. **Prototype** - Object cloning
13. **Singleton** - API client management
14. **Builder** - Complex object construction
15. **MVC** - Application architecture
16. **DAO** - Data access abstraction

Appendix B: Git References

Repository Information

GitHub Repository: https://github.com/Hemalv02/filr

Commit References:

- **Before Refactoring:** `0e3a19667017184fb1553df4b974ade3108ae48a`
  - Branch: main
  - Date: October 2024
  - Files: 12
  - LOC: 2,340
- **After Refactoring:** `74987d076ca4ea45377b9539b347474dfc6ec955`
  - Branch: sh-refactor
  - Date: November 2024
  - Files: 28
  - LOC: 3,125

Viewing Diff

To view the complete diff:

```
git diff 0e3a196670 74987d0766
```

To view specific file changes:

```
git diff 0e3a196670 74987d0766 -- src/lib/processors/DocumentProcessor.ts
```

References

1. Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley.

2. Freeman, E., & Robson, E. (2004). *Head First Design Patterns*. O'Reilly Media.

3. Martin, R. C. (2017). *Clean Architecture: A Craftsman's Guide to Software Structure and Design*. Prentice Hall.

4. Martin, R. C. (2008). *Clean Code: A Handbook of Agile Software Craftsmanship*. Prentice Hall.

5. Fowler, M. (2018). *Refactoring: Improving the Design of Existing Code* (2nd ed.). Addison-Wesley.